

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)			Risk – {damage cells / cause mutations / ionises cells} when handling (1) Don't accept causes cancer Control measure – limit exposure <u>time</u> / use tongs / shielding when not in use / source directed away from teacher [and student] (1) Don't accept wear gloves or goggles or use tweezers	2			2		2
	(b)	(i)		Paper has no effect on count rate Don't accept alpha is low penetrating or alpha is stopped by paper			1	1		1
		(ii)		<u>Count rate</u> drops at aluminium so beta present (1) <u>Count rate</u> drops [again] at lead so gamma present (1) Award 1 mark only for <u>count rate</u> drops at aluminium and lead Don't accept aluminium absorbs beta or lead absorbs gamma without reference to count rate			2	2		2
		(iii)	I	Radon / cosmic rays / rocks / food and drink / buildings / nuclear power stations / medical uses Accept Sun or stars Don't accept X-rays or Big Bang or CMBR	1			1		1
			II	[Measure the background radiation and] subtract background from all readings Accept a numerical value taken away	1			1		1
				Question 1 total	4	0	3	7	0	6